link-state routing

will keep idea of sharing state with neighbors…

but weren't sharing enough state!

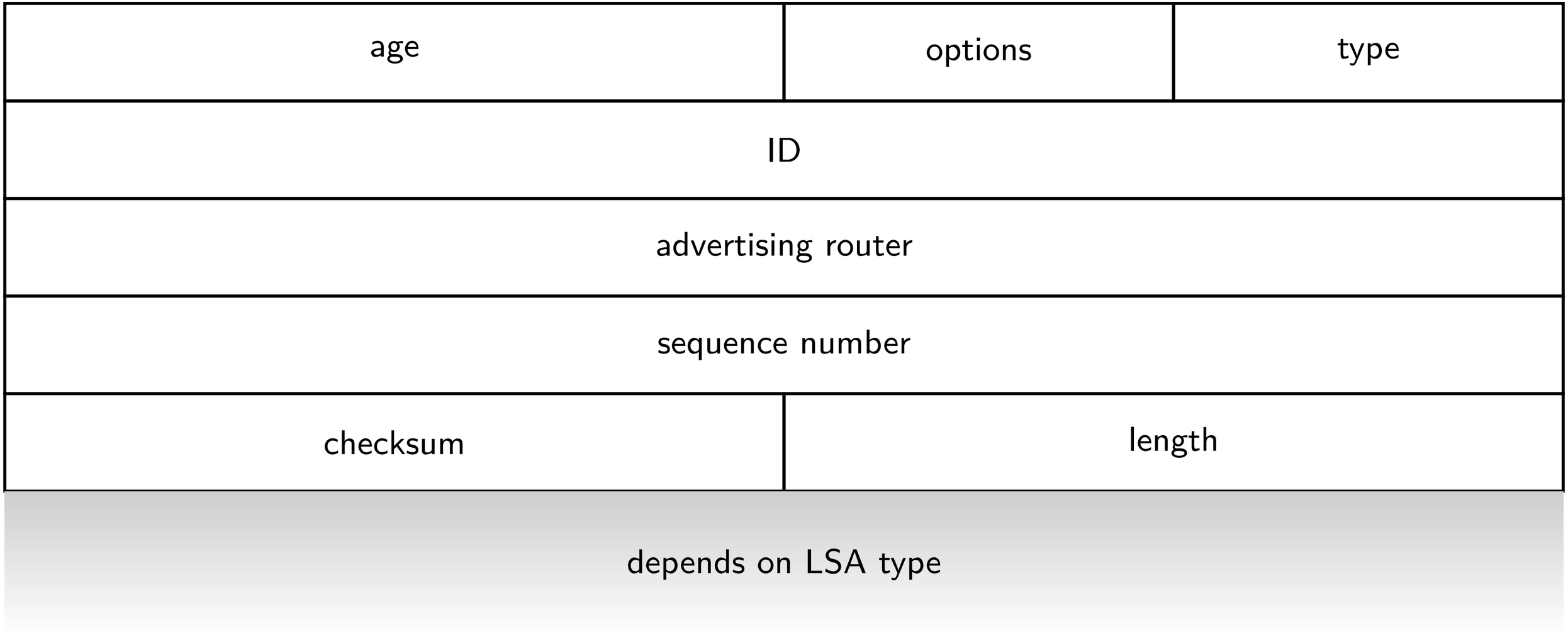
other routing idea:

routers collect *complete map* of network

example protocol for this: OSPF

Open Shortest Path First

OSPF link-state advertisements (router)



OSPF LSA sequences/ages

sequence number for getting correction version of LSAs

- some tricky rules to handle routers restarting (losing track of sequence number) and sequence number wraparound

maximum ‘age’ for link-state advertisements

- typically minutes
- too-old LSAs not used for routing
- deliberately setting age = MaxAge used to invalidate LSAs

OSPF LSA types

‘router’:

- list of links for router
- links = connect to other router or network
- links refer to ID numbers of network/router LSAs
- metrics for each link

‘network’:

- list of routers for network
- different version of external and internal networks

(later) ‘summary’:

- part of support for *areas*
- used when sysadmin doesn’t want all routers processing whole network map

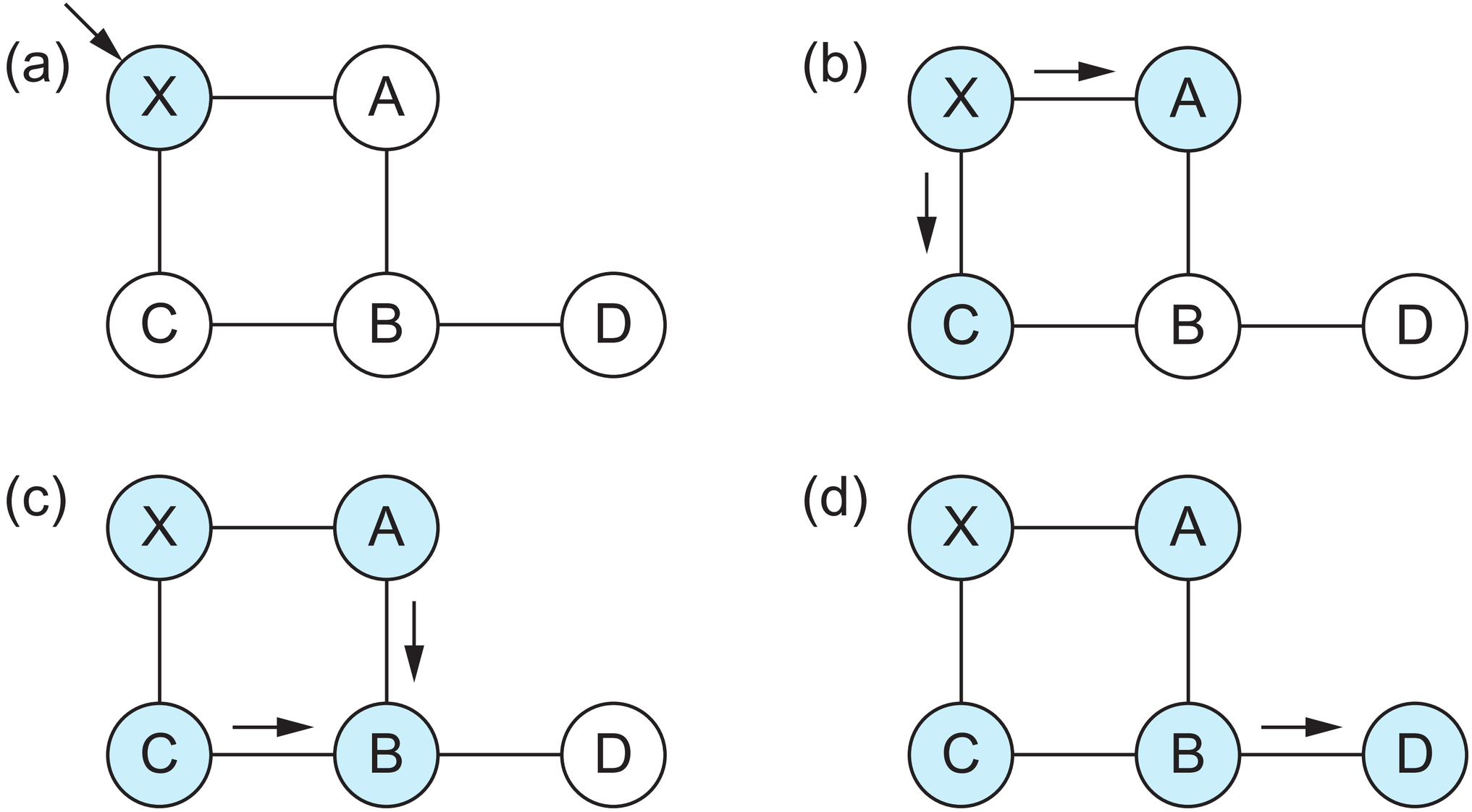
link-state database

whole collection of advertisements

every router, network, link between router+network

all the metrics for those

reliable flooding (picture)



Peterson and Davie, *Computer Networks: A Systems Approach*, Figure 88

missing from the picture

in picture: seems like each router directly connected to each other

often we have multiple routers connected to local network

can/will share link state packets by broadcasting on local network

reliable flooding in OSPF — setup

for each subnetwork:

- choose a designated and backup router
- make sure backup becomes designated on failure

designated router will take care of propogating updates to everyone on network

… including waiting for acknowledgments, etc.

reliable flooding in OSPF

then, when receiving/generating link state packet:

send to every designed+backup router of subnetwork that

- you are connected to, and
- you are not designated/backup router for, and
- you did not receive the packet from

send to every router on every subnetwork that

- you are designated router for

send = send + resend if no ACK

finding shortest paths

given full picture of network

want to find all shortest paths from self

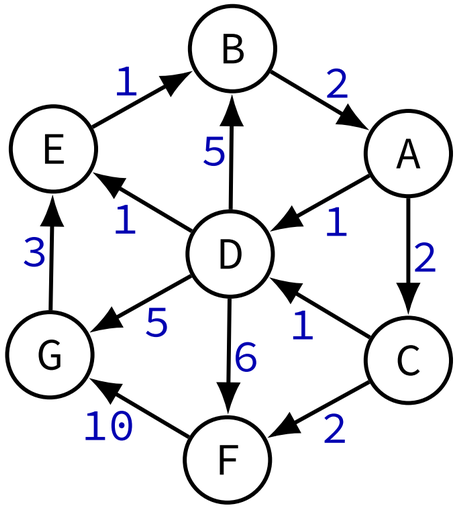
shortest ‘distance’ = lowest sum of metric

only need next hop, but will compute whole path to find that

assumption: everyone using shortest path

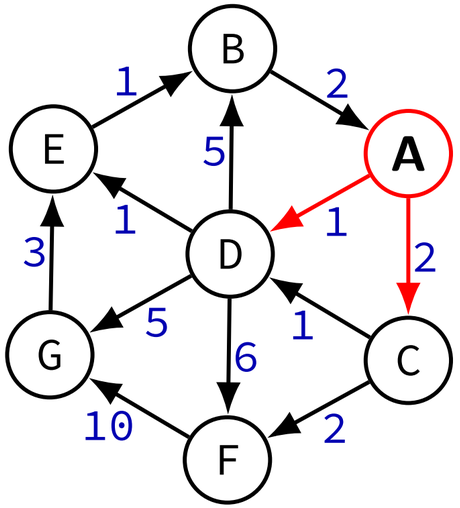
usual solution: Dijkstra’s algorithm

Dijkstra's algorithm example 1



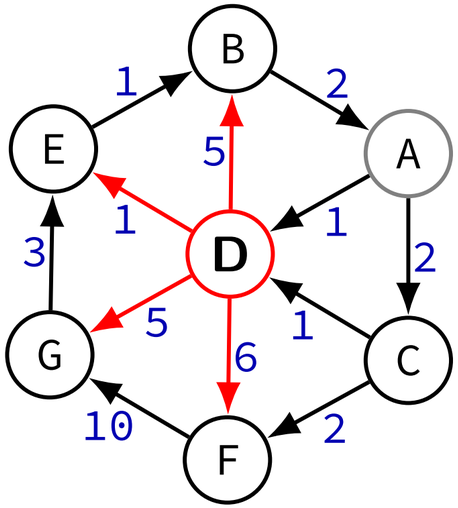
	dist	prev	path
A	0	—	A
B	∞	—	—
C	∞	—	—
D	∞	—	—
E	∞	—	—
F	∞	—	—
G	∞	—	—

Dijkstra's algorithm example 1



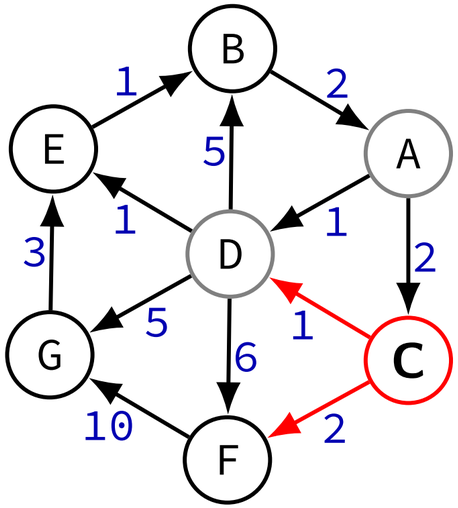
	dist	prev	path
A	0	—	A
B	∞	—	—
C	2	A	A→C
D	1	A	A→D
E	∞	—	—
F	∞	—	—
G	∞	—	—

Dijkstra's algorithm example 1



	dist	prev	path
A	0	—	A
B	6	D	A→D→B
C	2	A	A→C
D	1	A	A→D
E	2	D	A→D→E
F	7	D	A→D→F
G	6	D	A→D→G

Dijkstra's algorithm example 1

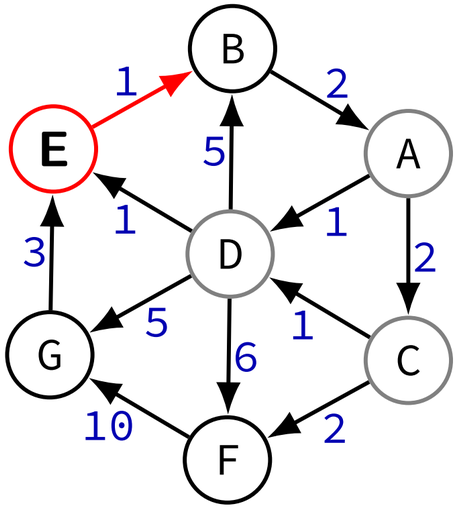


D is adjacent —
but not a shorter path

	dist	prev	path
A	0	—	A
B	6	D	A→D→B
C	2	A	A→C
D	1	A	A→D
E	2	D	A→D→E
F	4	C	A→C→F
G	6	D	A→D→G

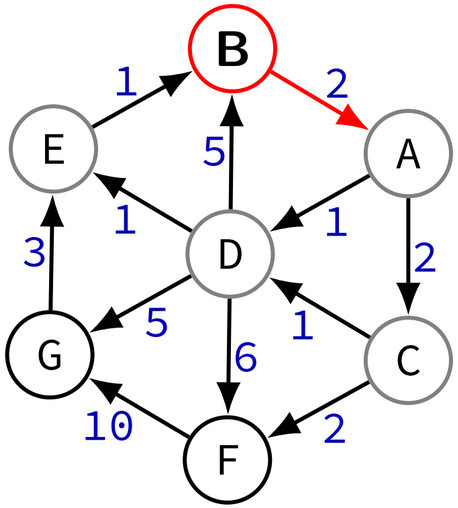
F updated from distance 7 (via D)
to distance 4 (via C)

Dijkstra's algorithm example 1



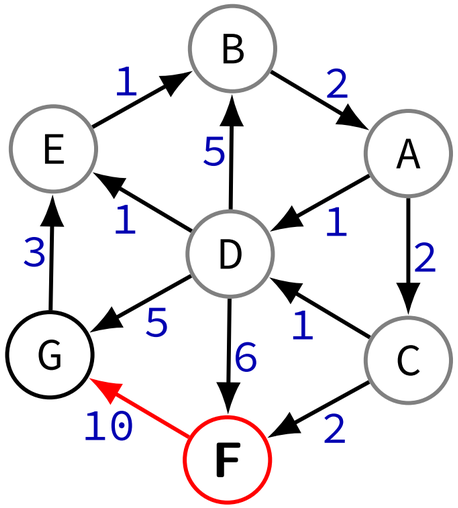
	dist	prev	path
A	0	—	A
B	3	E	A→D→E→B
C	2	A	A→C
D	1	A	A→D
E	2	D	A→D→E
F	4	C	A→C→F
G	6	D	A→D→G

Dijkstra's algorithm example 1



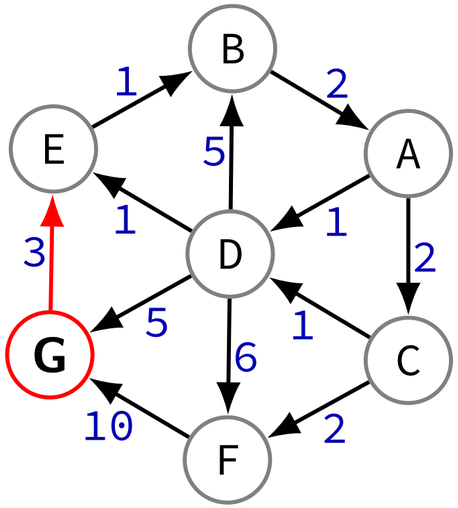
	dist	prev	path
A	0	—	A
B	3	E	A→D→E→B
C	2	A	A→C
D	1	A	A→D
E	2	D	A→D→E
F	4	C	A→C→F
G	6	D	A→D→G

Dijkstra's algorithm example 1



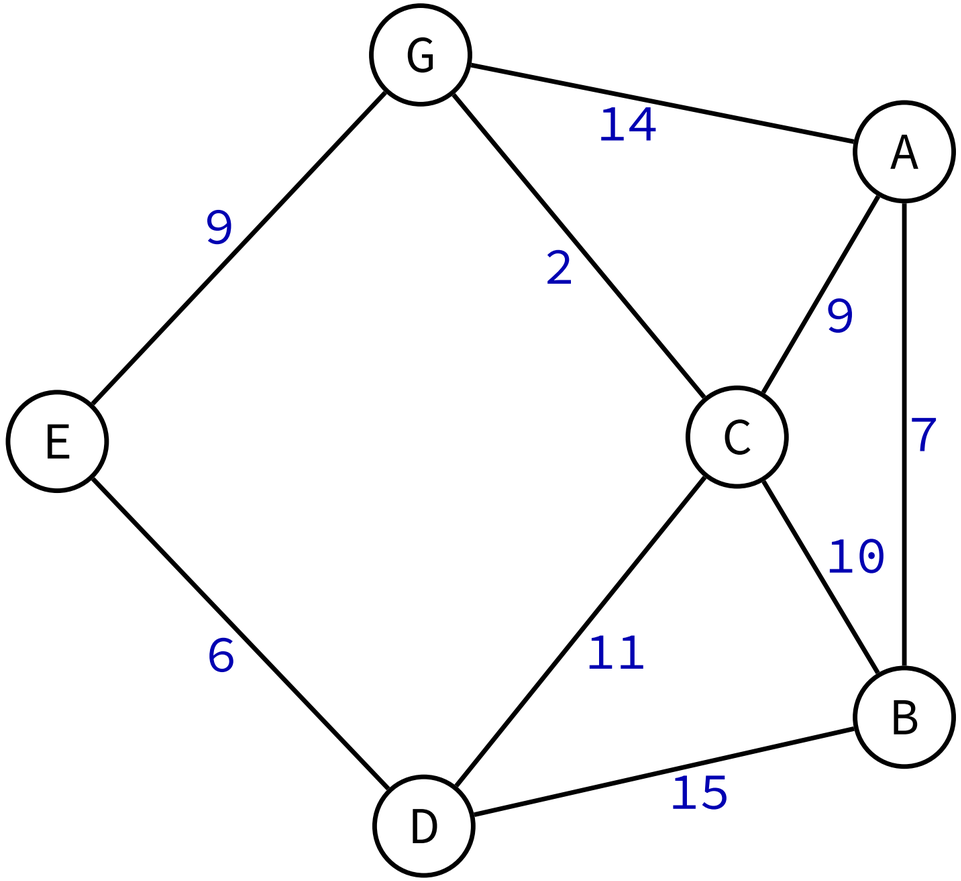
	dist	prev	path
A	0	—	A
B	3	E	A→D→E→B
C	2	A	A→C
D	1	A	A→D
E	2	D	A→D→E
F	4	C	A→C→F
G	6	D	A→D→G

Dijkstra's algorithm example 1



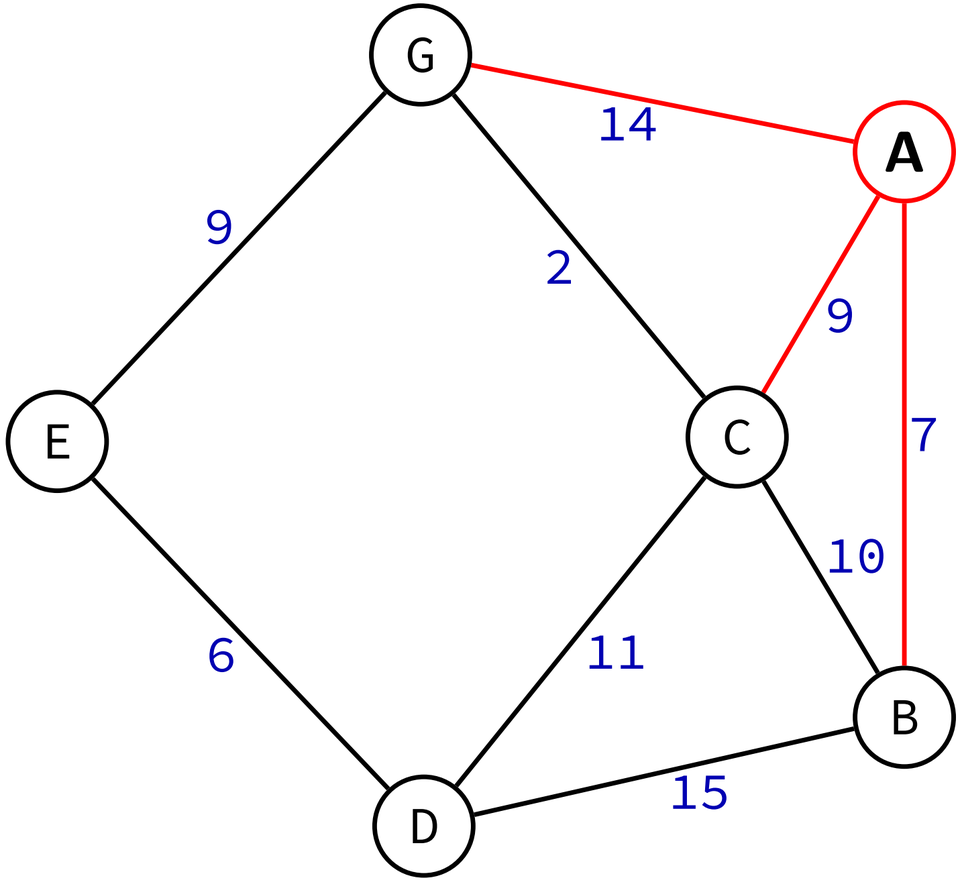
	dist	prev	path
A	0	—	A
B	3	E	A→D→E→B
C	2	A	A→C
D	1	A	A→D
E	2	D	A→D→E
F	4	C	A→C→F
G	6	D	A→D→G

Dijkstra's algorithm example 2



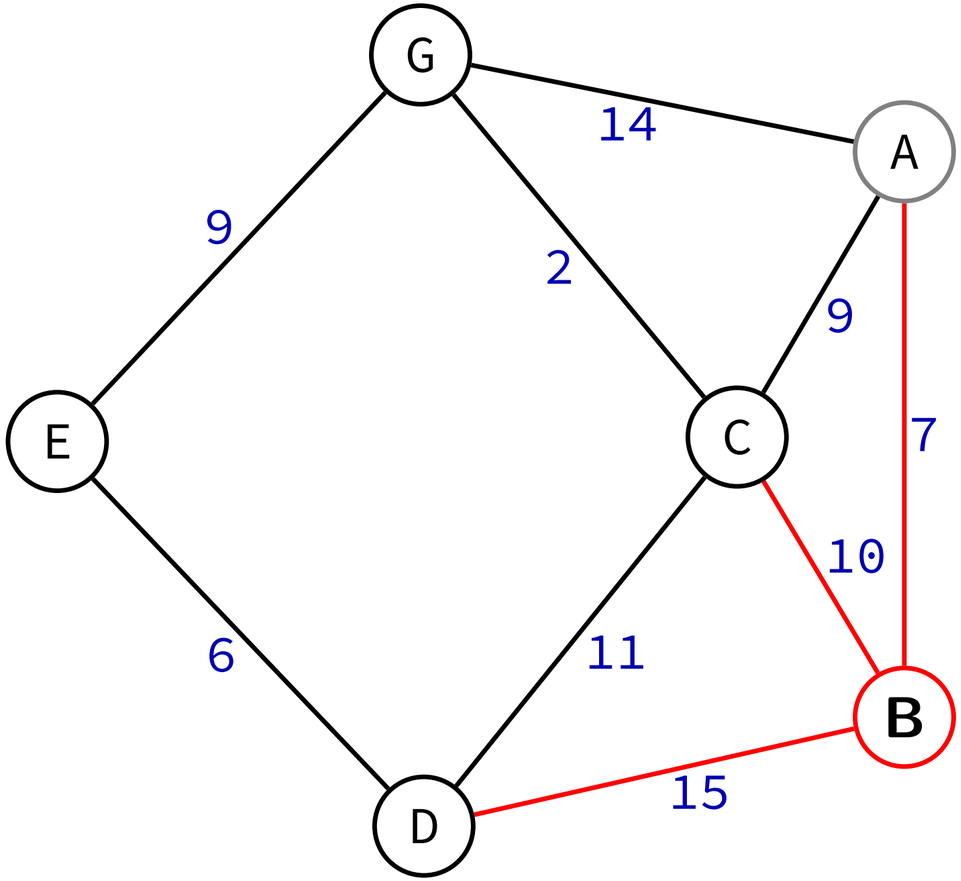
	dist	prev	path
A	0	—	A
B	∞	—	—
C	∞	—	—
D	∞	—	—
E	∞	—	—
G	∞	—	—

Dijkstra's algorithm example 2



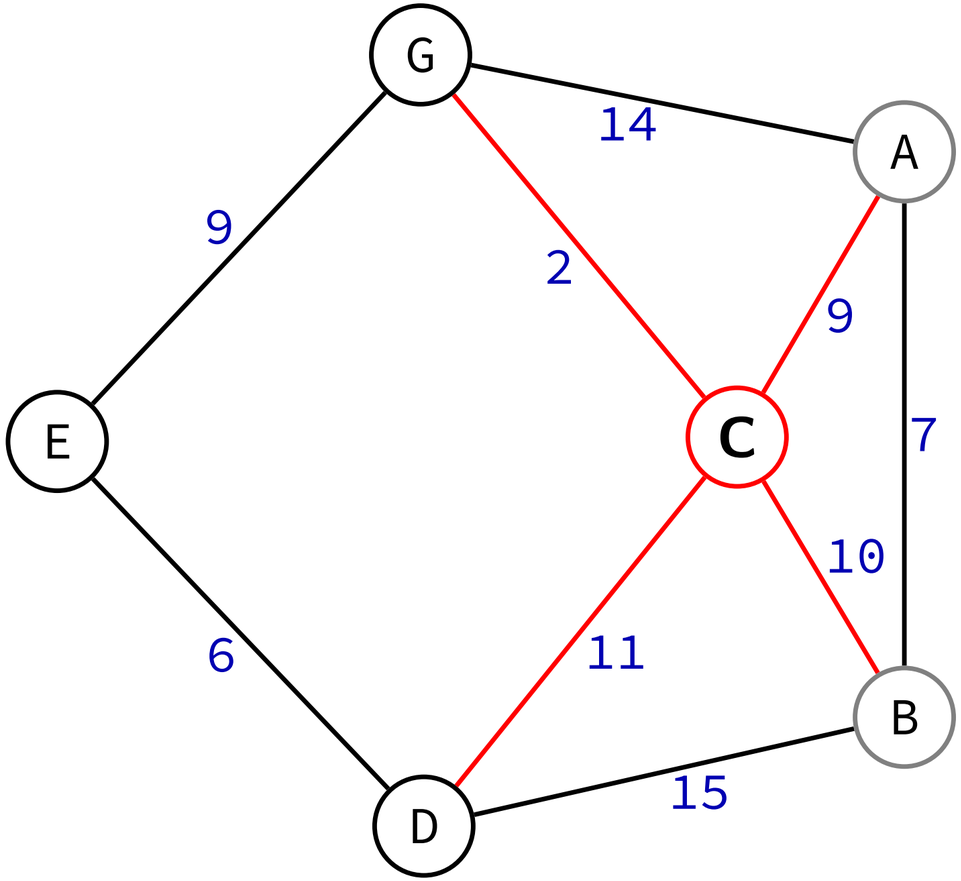
	dist	prev	path
A	0	—	A
B	7	A	A→B
C	9	A	A→C
D	∞	—	—
E	∞	—	—
G	14	A	A→G

Dijkstra's algorithm example 2



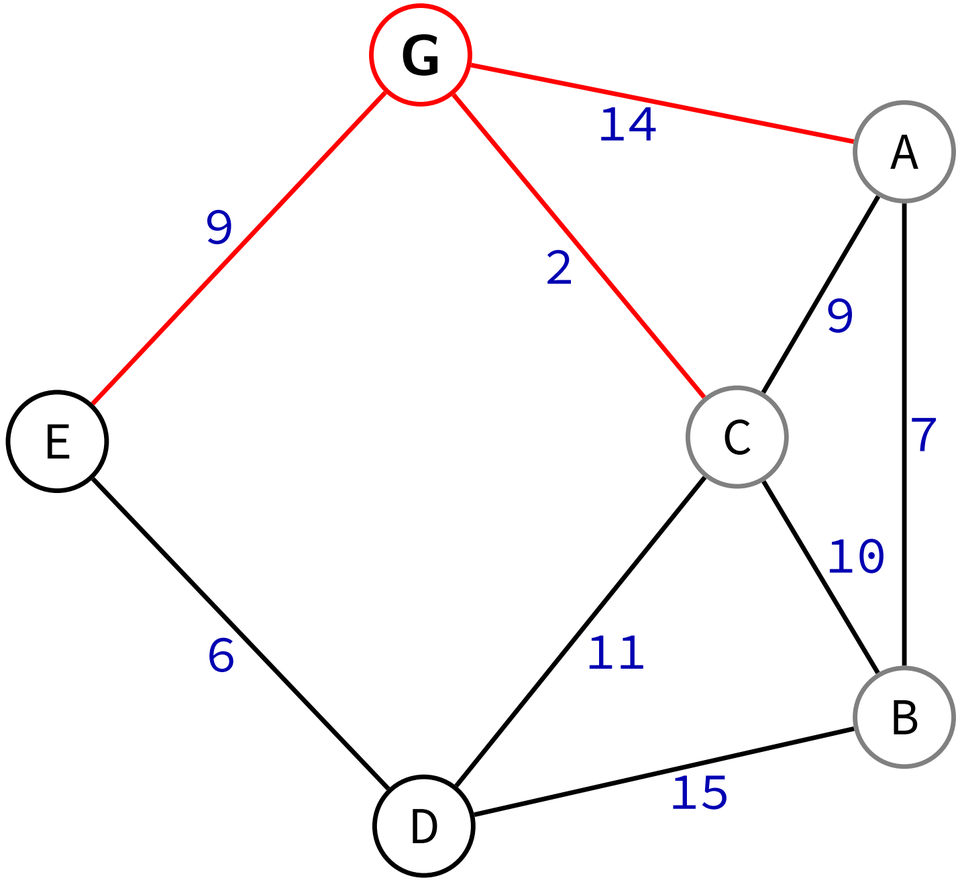
	dist	prev	path
A	0	—	A
B	7	A	A→B
C	9	A	A→C
D	22	B	A→B→D
E	∞	—	—
G	14	A	A→G

Dijkstra's algorithm example 2



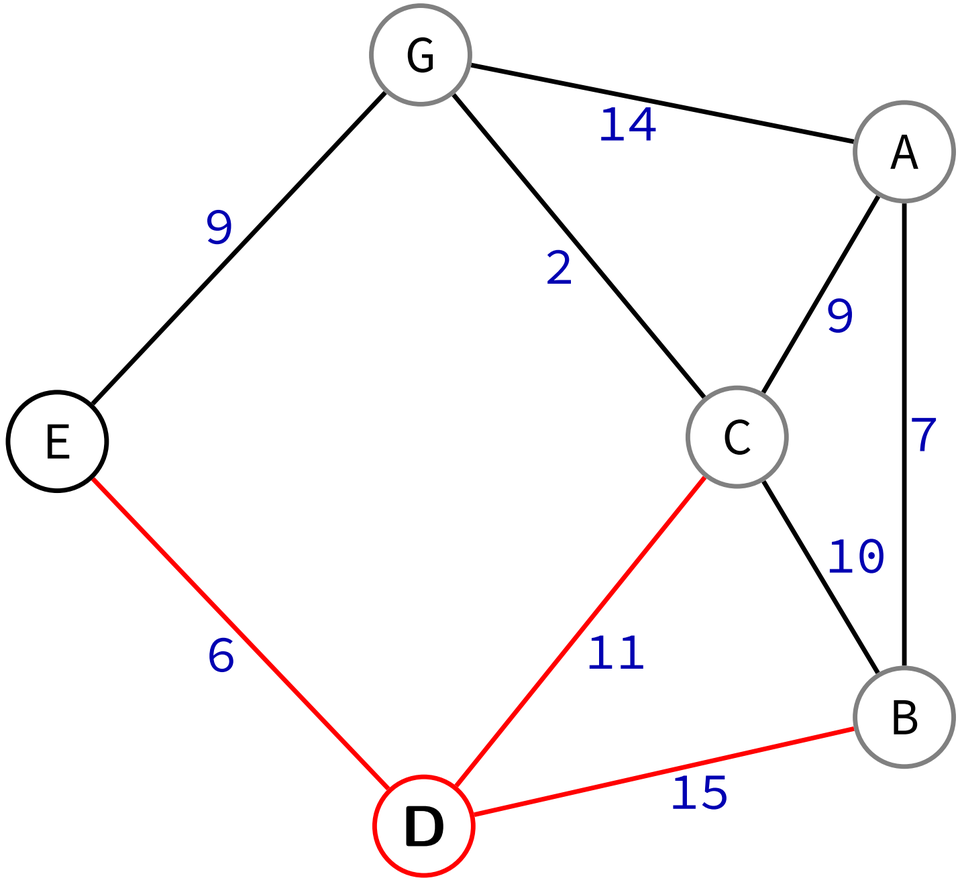
	dist	prev	path
A	0	—	A
B	7	A	A→B
C	9	A	A→C
D	20	C	A→C→D
E	∞	—	—
G	11	C	A→C→G

Dijkstra's algorithm example 2



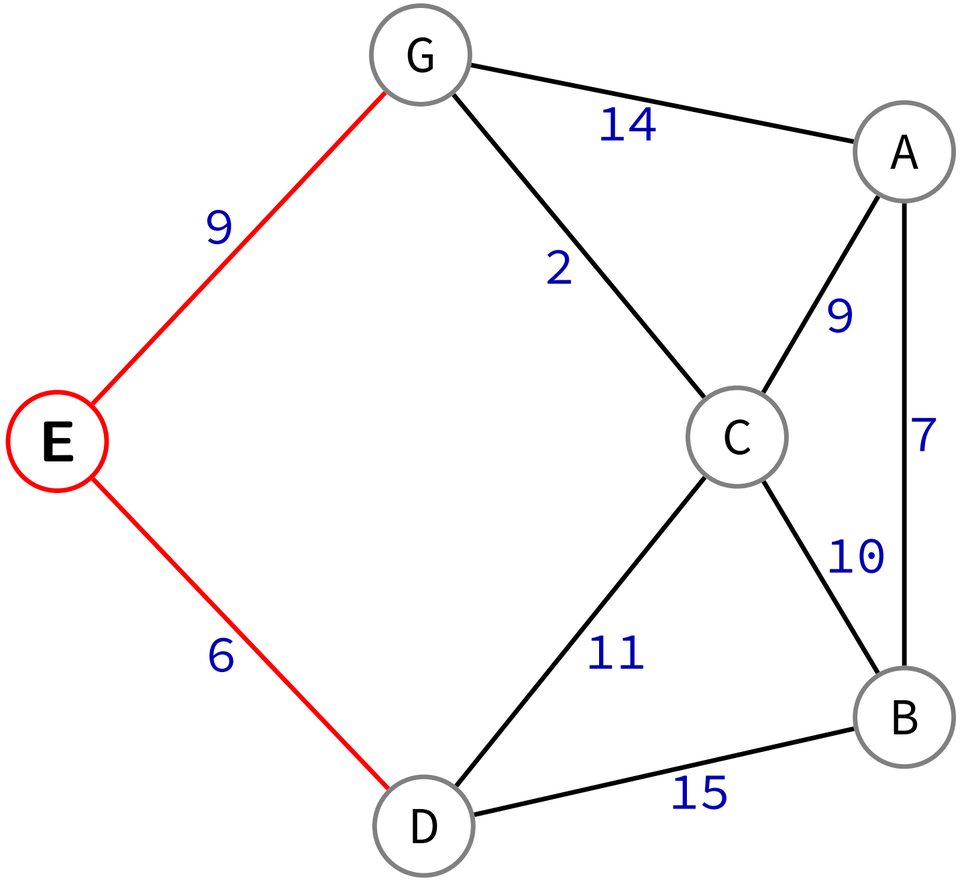
	dist	prev	path
A	0	—	A
B	7	A	A→B
C	9	A	A→C
D	20	C	A→C→D
E	20	G	A→C→G→E
G	11	C	A→C→G

Dijkstra's algorithm example 2



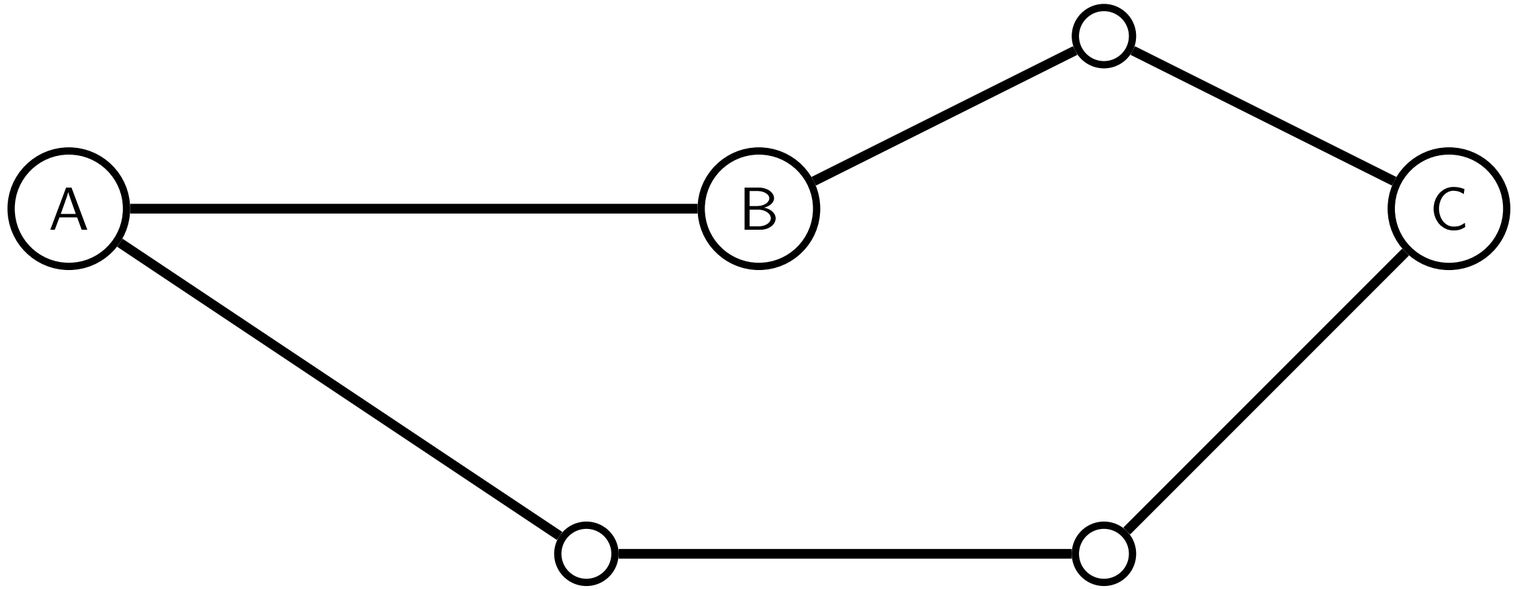
	dist	prev	path
A	0	—	A
B	7	A	A→B
C	9	A	A→C
D	20	C	A→C→D
E	20	G	A→C→G→E
G	11	C	A→C→G

Dijkstra's algorithm example 2



	dist	prev	path
A	0	—	A
B	7	A	A→B
C	9	A	A→C
D	20	C	A→C→D
E	20	G	A→C→G→E
G	11	C	A→C→G

consistency

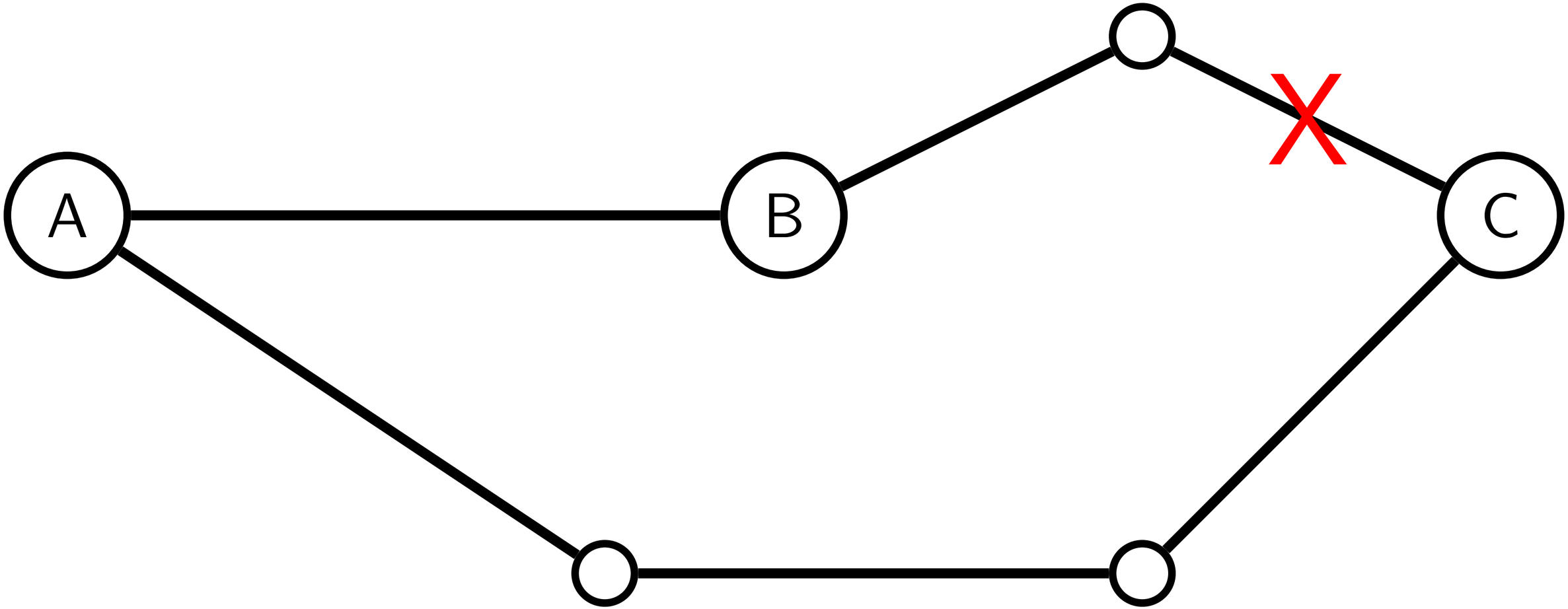


if A sends packet for C to B, how does A know B won't send it back?

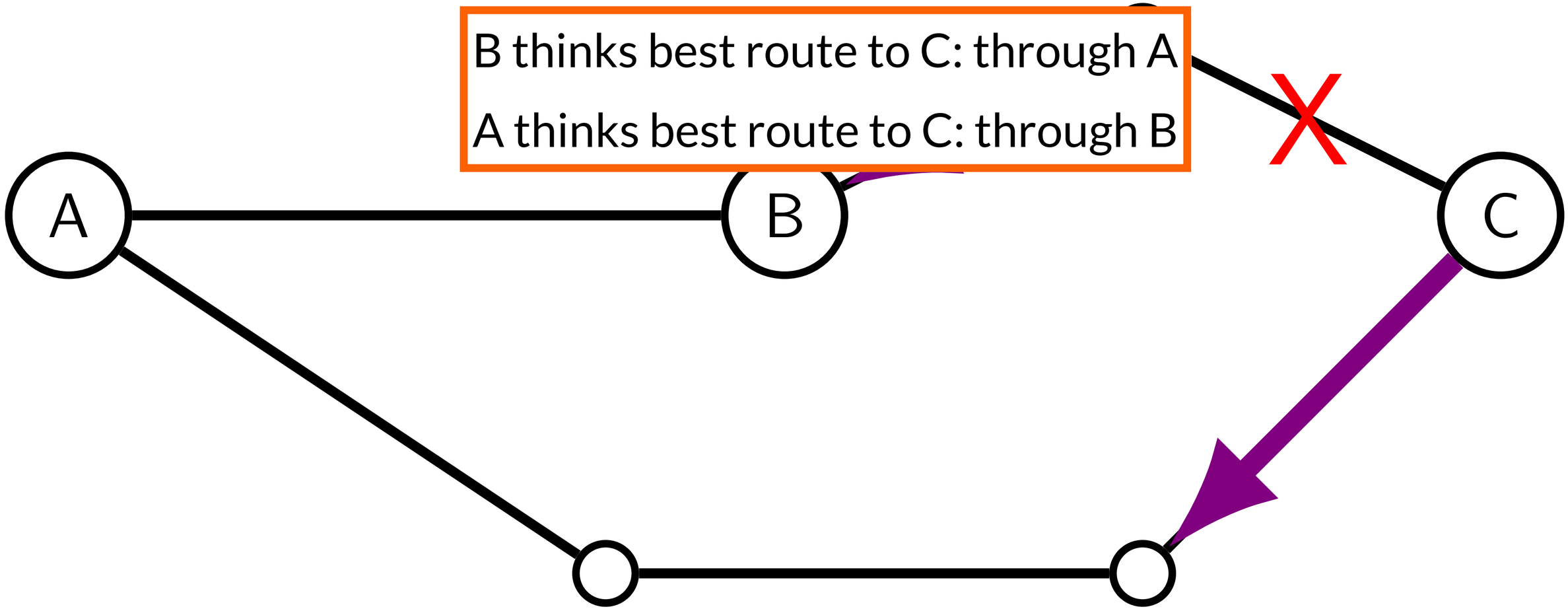
hope: if A thought shortest path to C was through B, then B should agree

problem: not always true

inconsistency



inconsistency



temporary bad routes

while waiting for link state updates to propogate

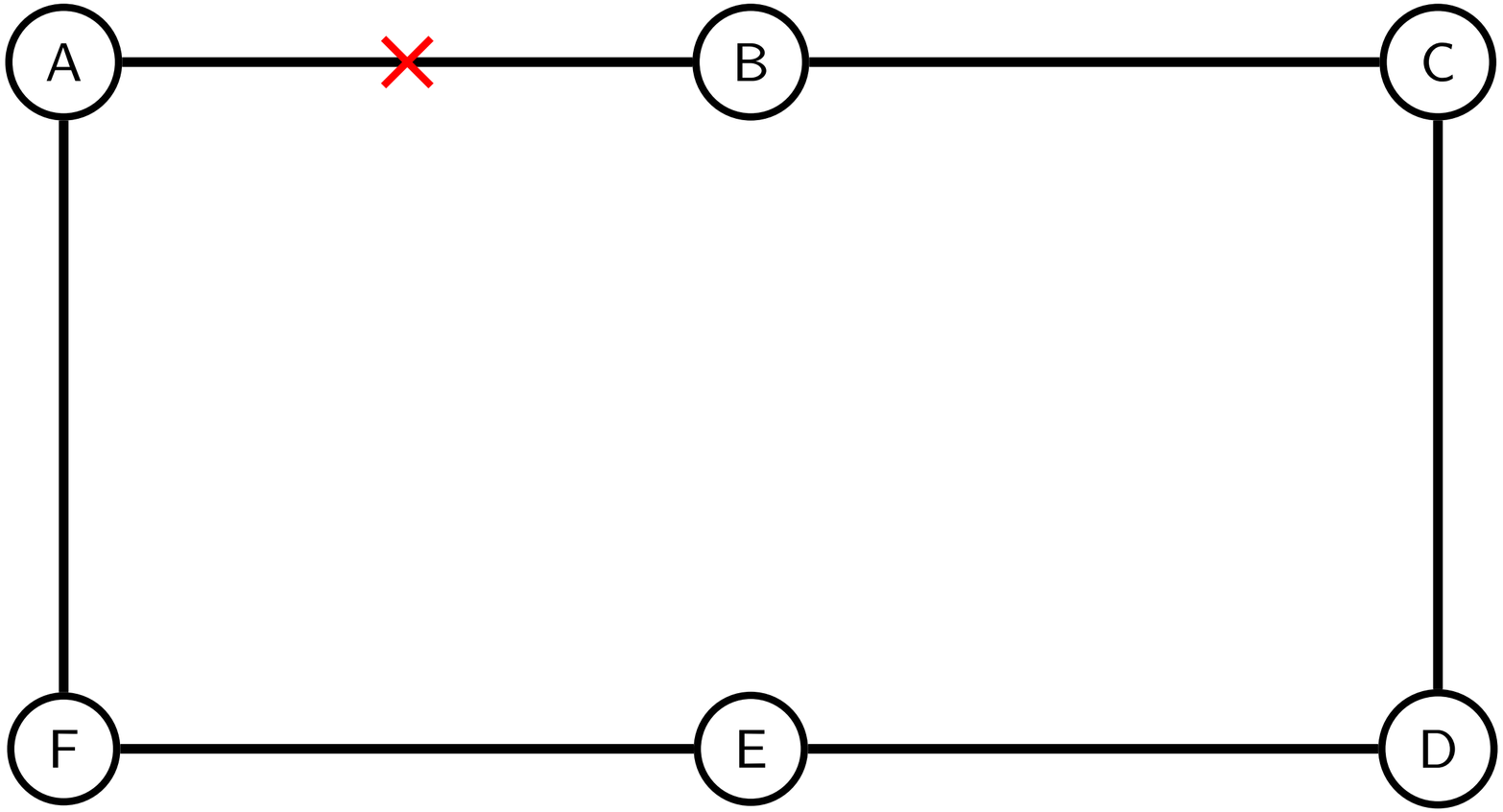
can have too-slow routes

can have routing loops

hope: this is only a few seconds at most

- and routing loop doesn't cause huge explosion of traffic

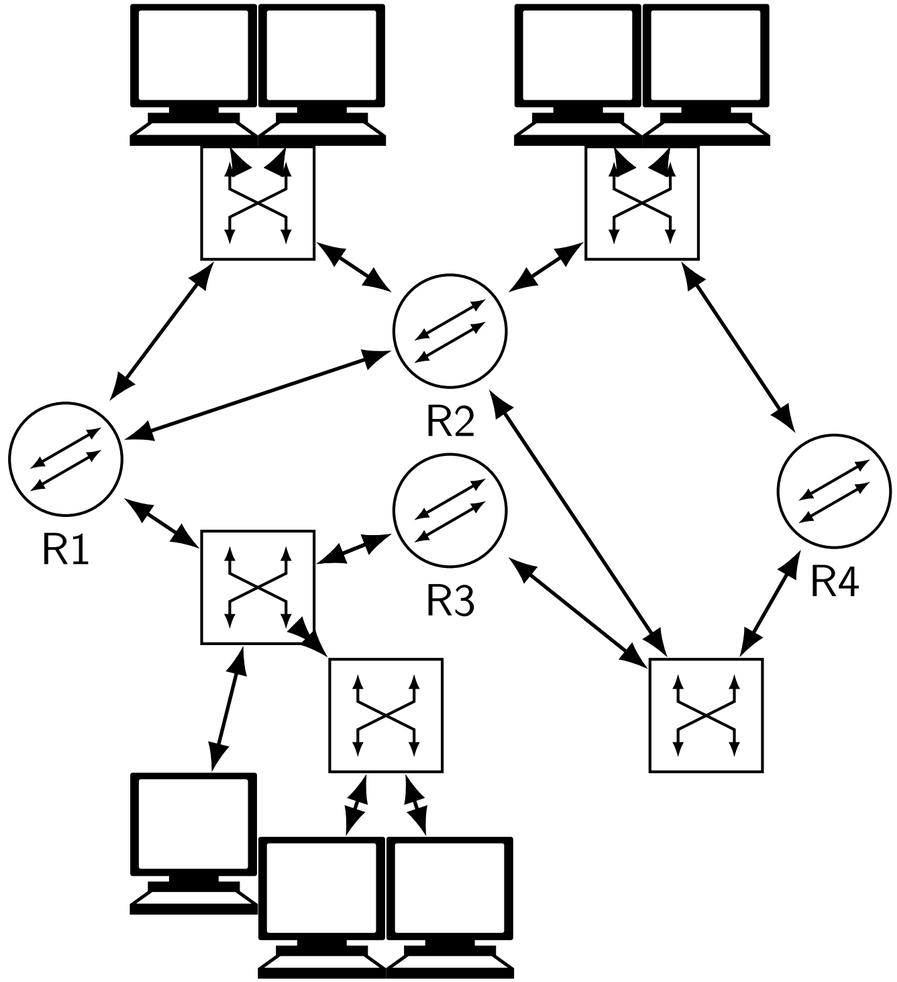
convergence time



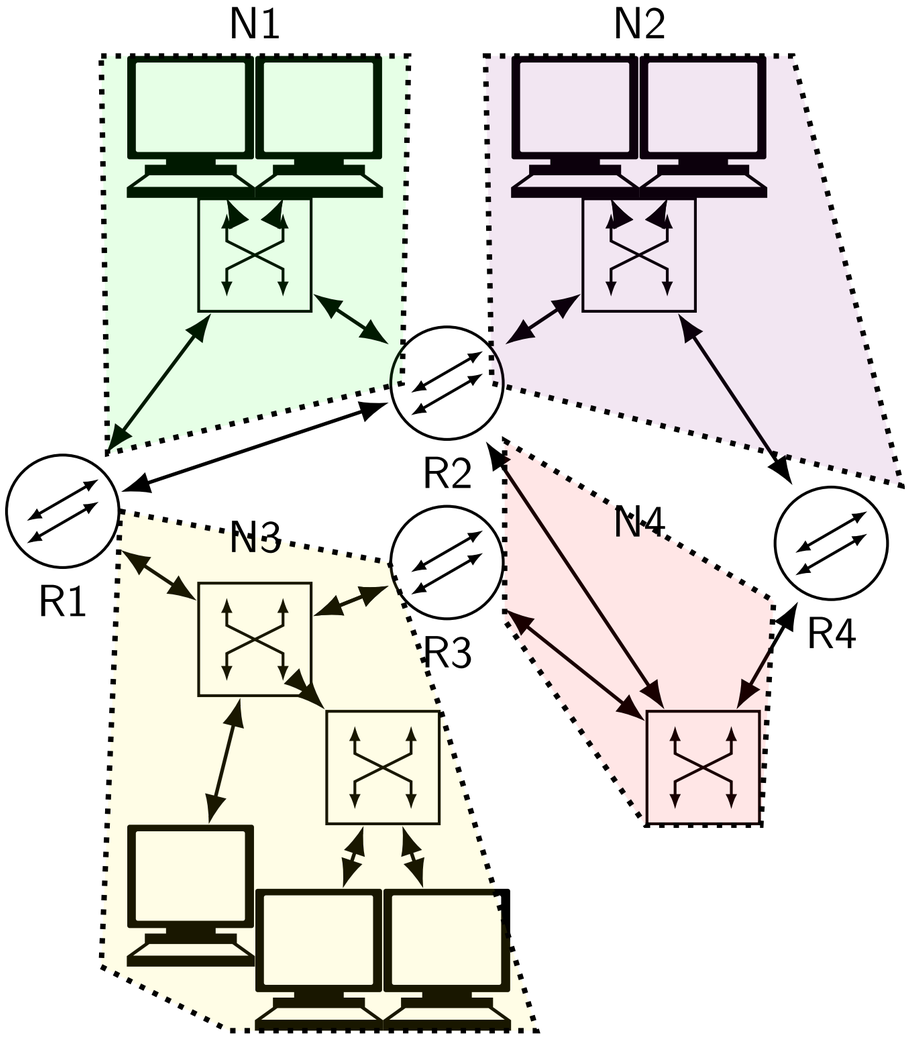
exercise: how many steps to fix A's next hop for B, C, D, etc....

- with distance vector?
- with link state?

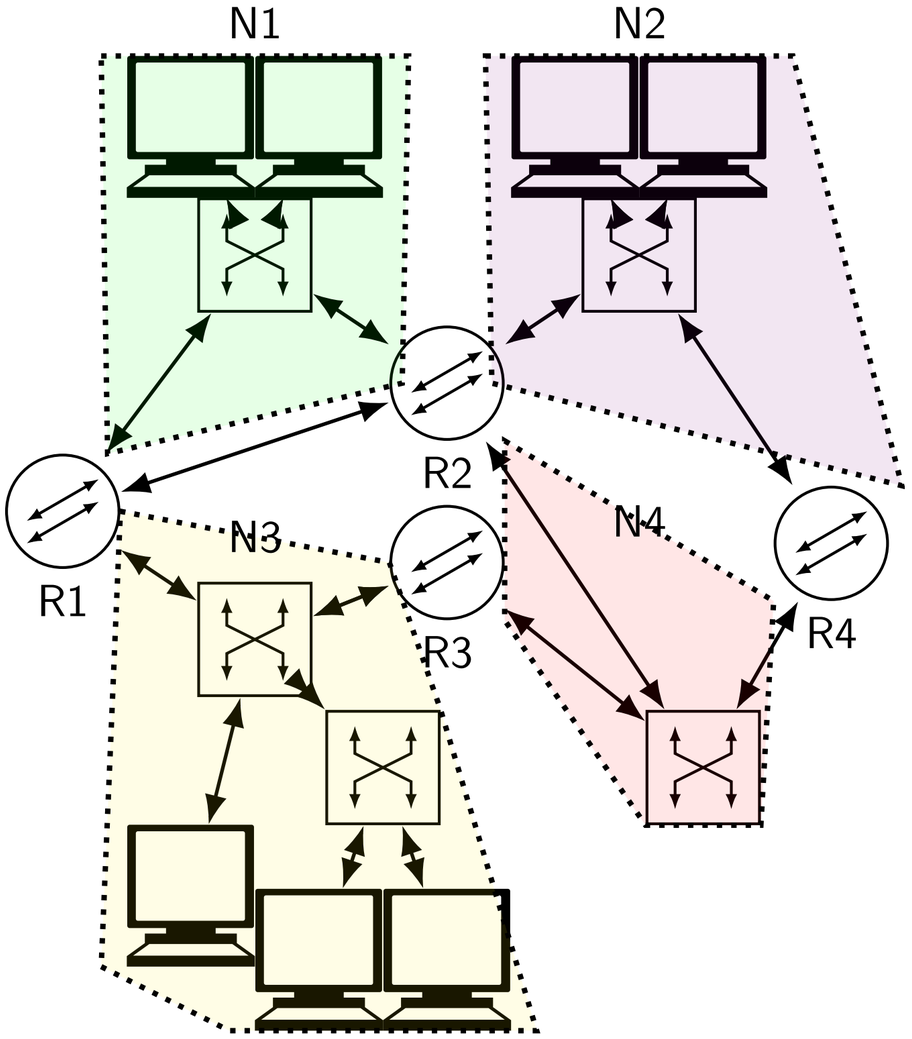
networks v routers (LS)

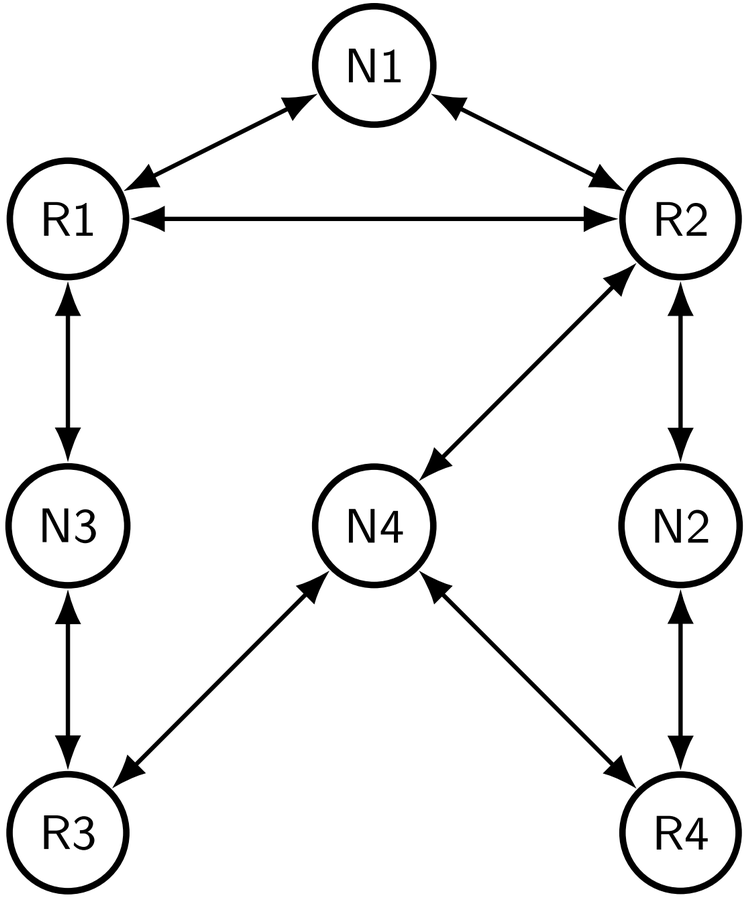


networks v routers (LS)

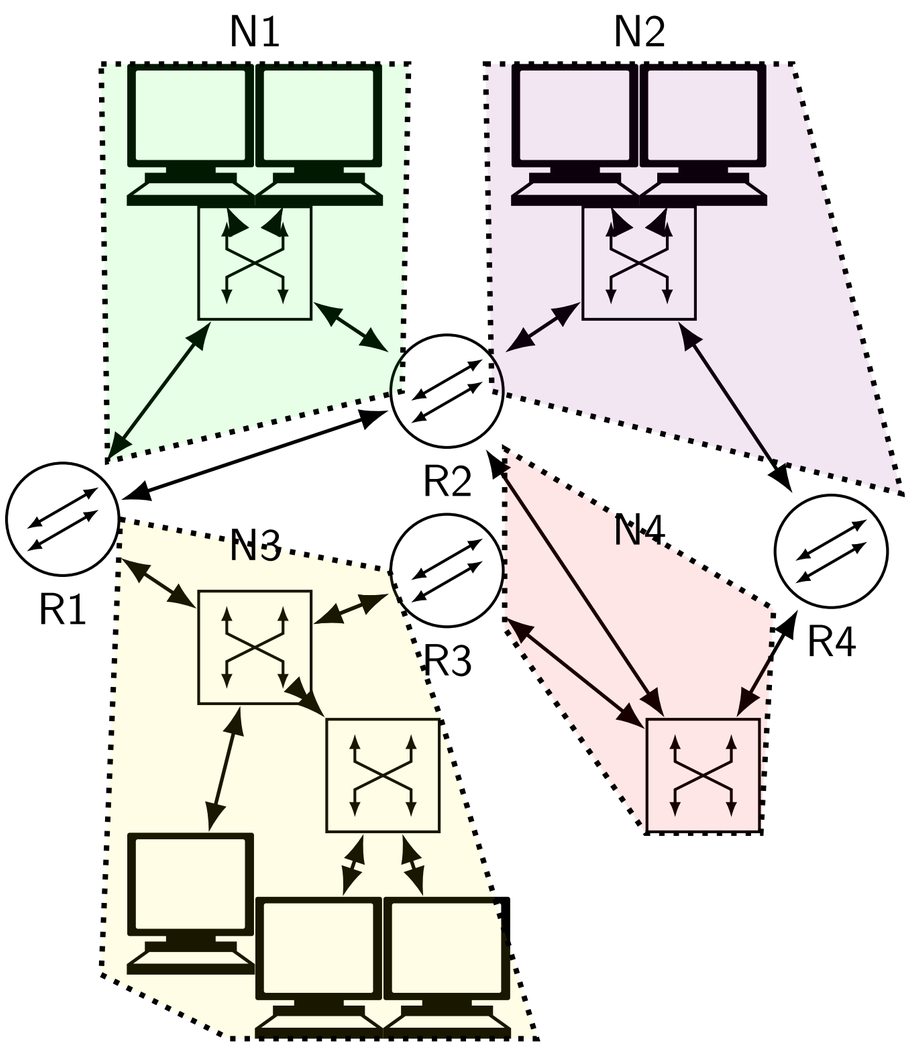


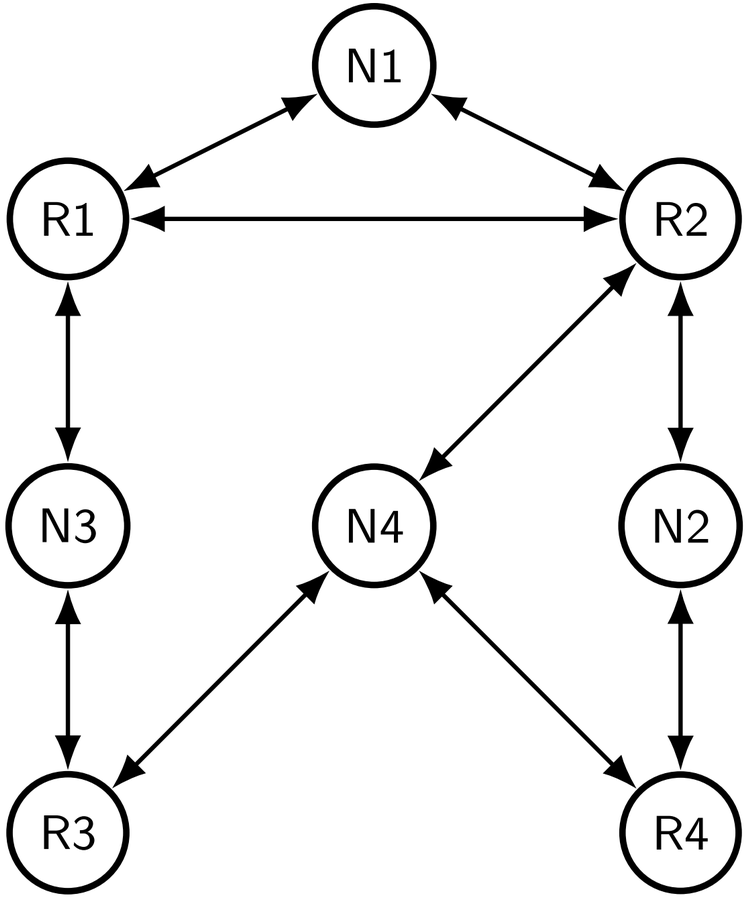
networks v routers (LS)





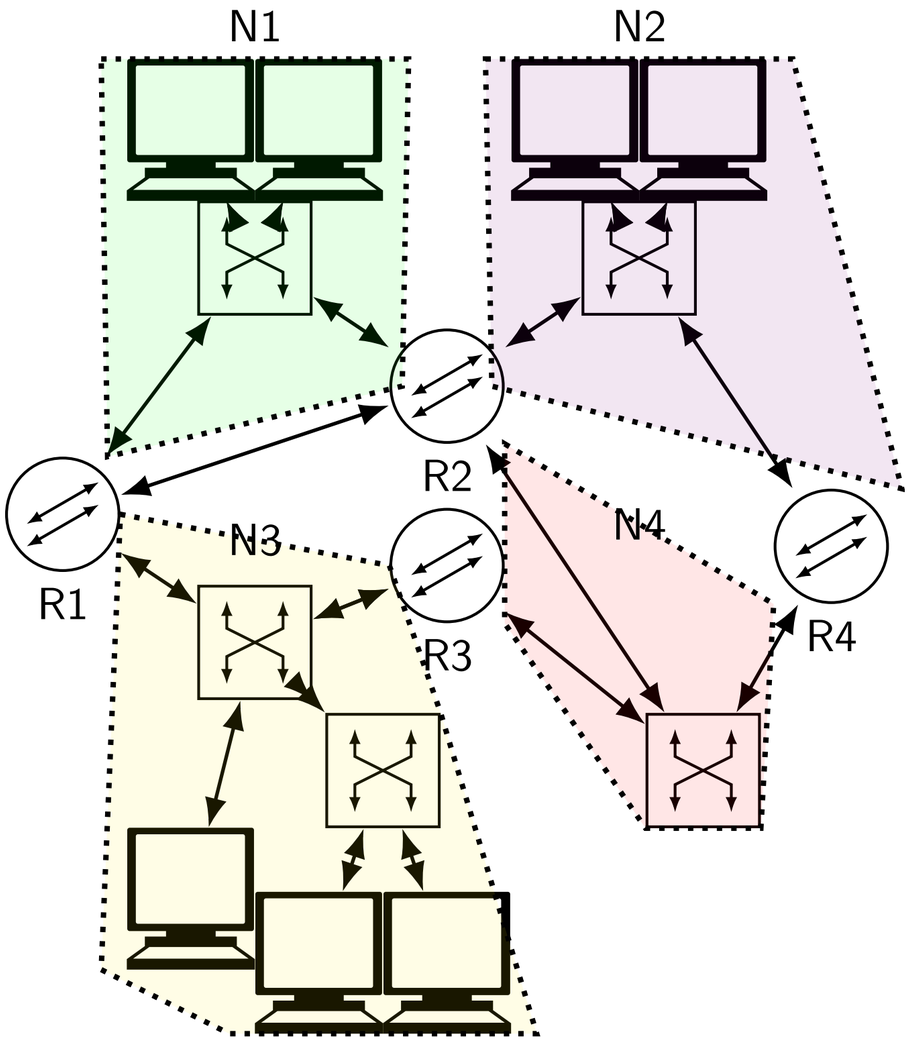
networks v routers (LS)

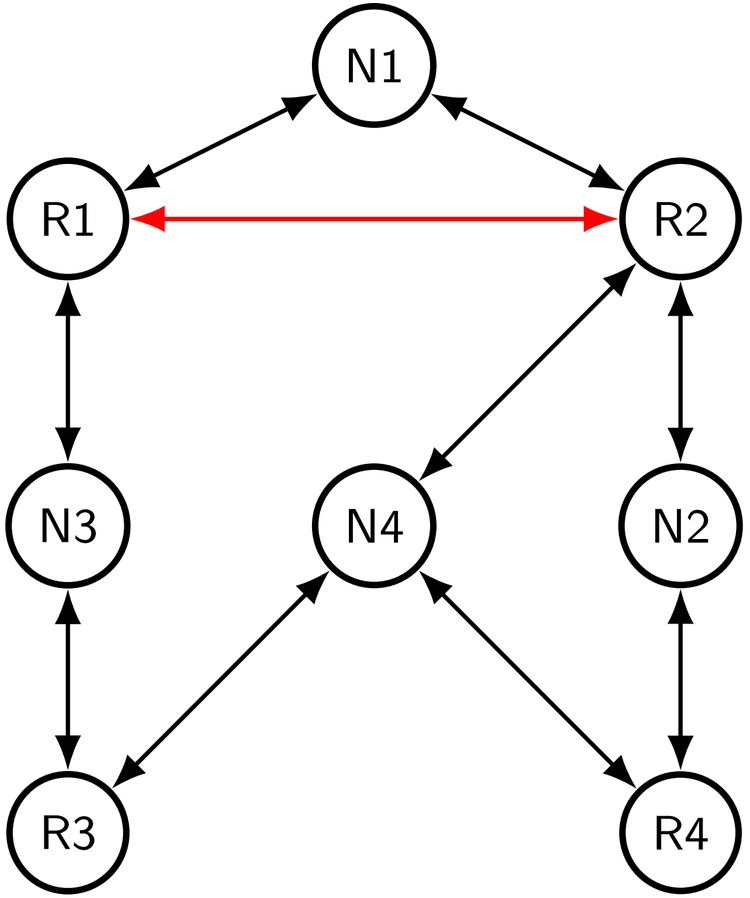




graph has nodes for routers+networks

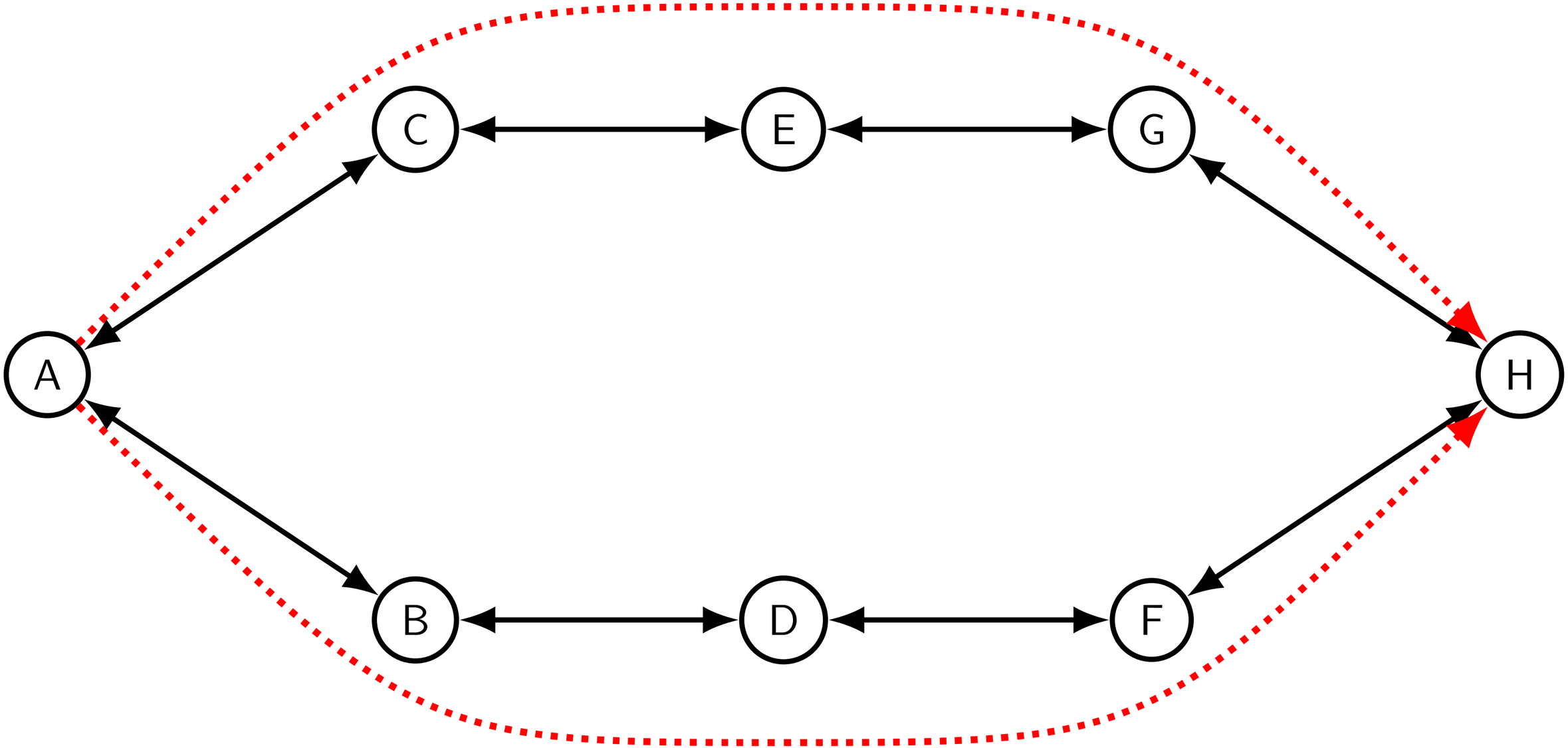
networks v routers (LS)





can have direct router-router link

two good choices?



splitting packets

naïve idea: send every other packet on bottom link

problem: bottom and top link will have different latencies

(even if only temporarily from queuing)

$\Longrightarrow$ packets will be reordered a lot

this is pretty bad for TCP

(and many other things)

equal cost multipath (ECMP)

split packets by flow

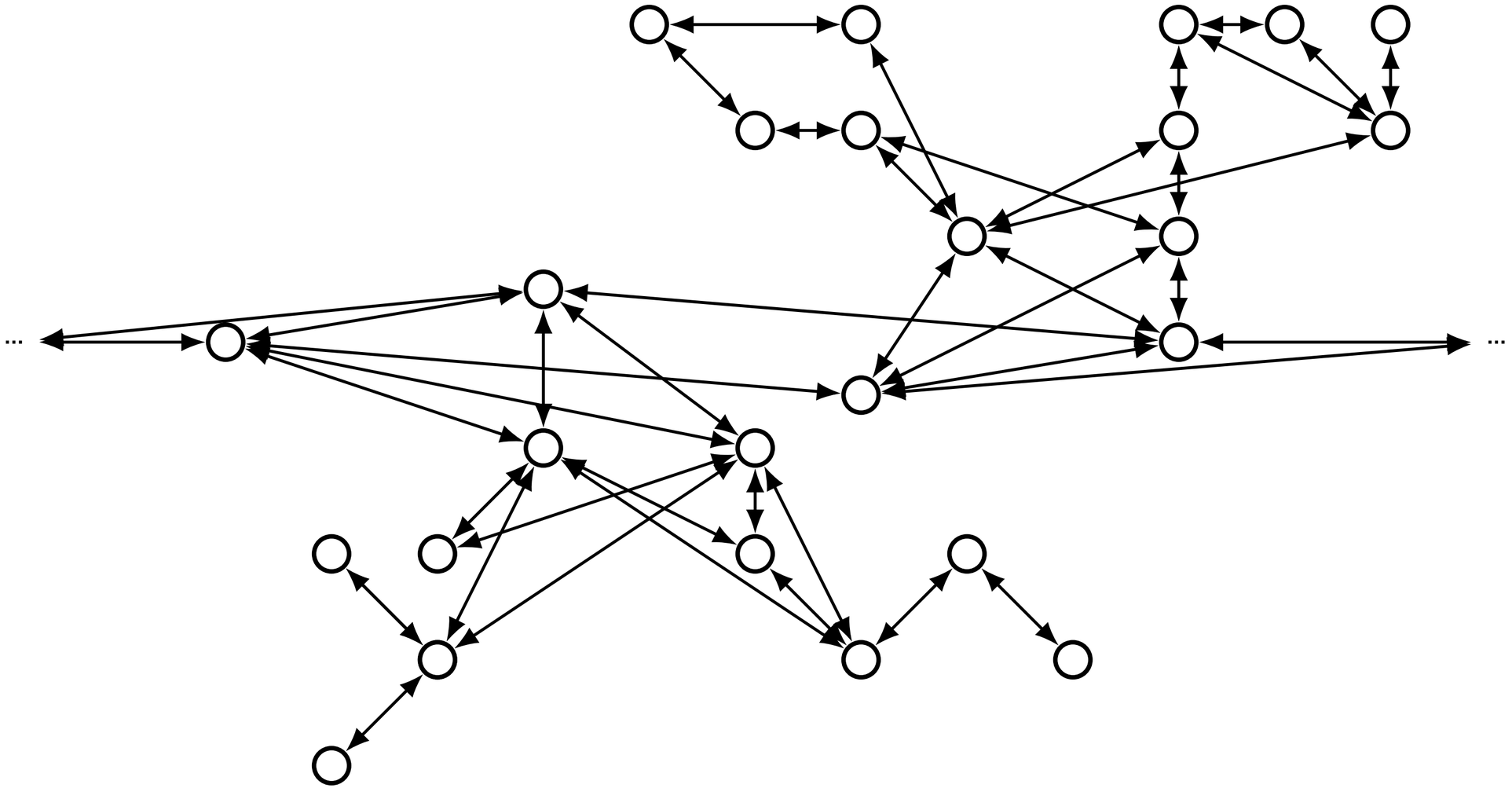
goal:

- each TCP connection chooses one of the N links
- … but don't want to track list of TCP connections

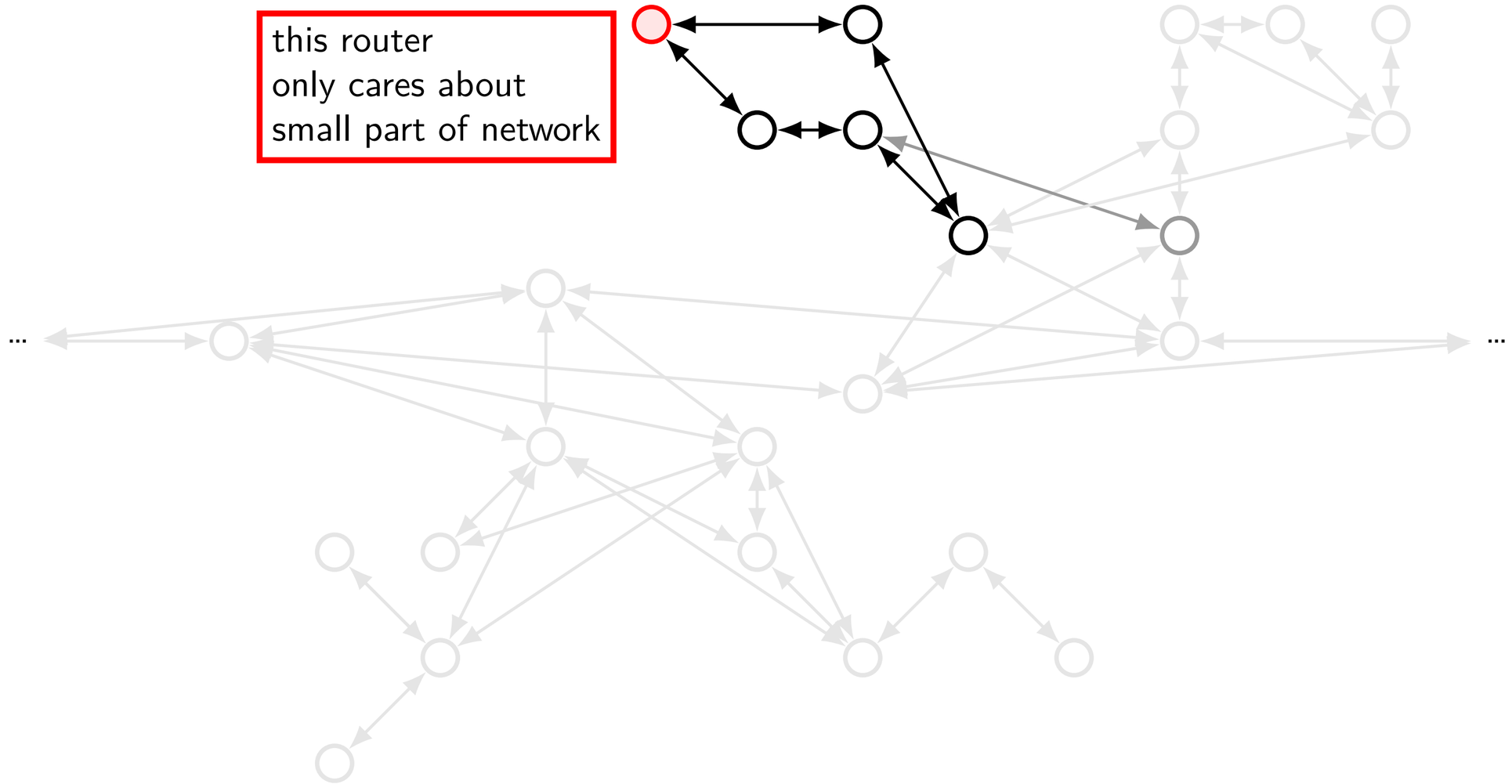
solution:

- take a hash of the connection info in header
- use link index $\left\lfloor \frac{\text{hash value} \times N}{\text{max hash value}} \right\rfloor$

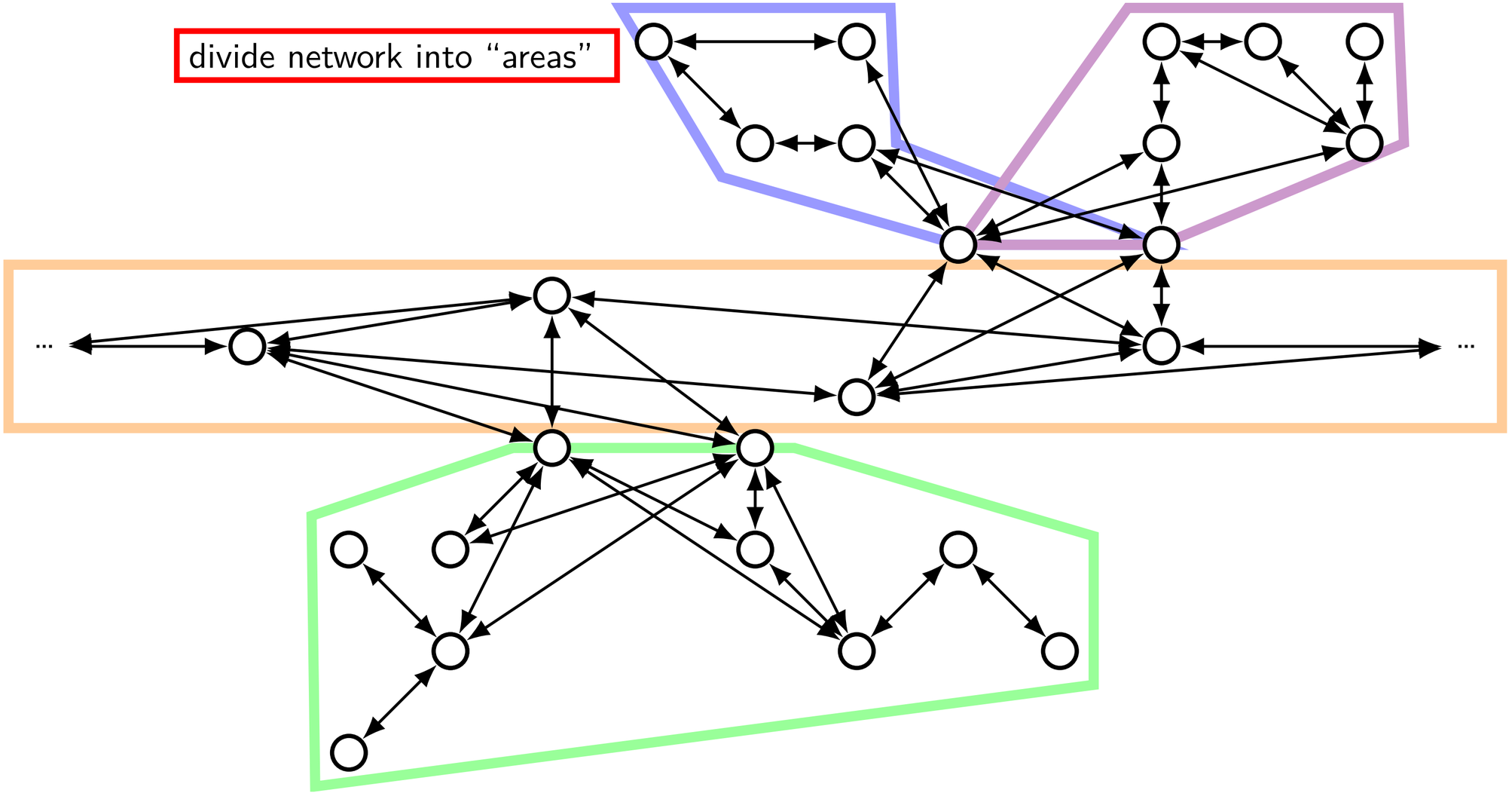
a big network



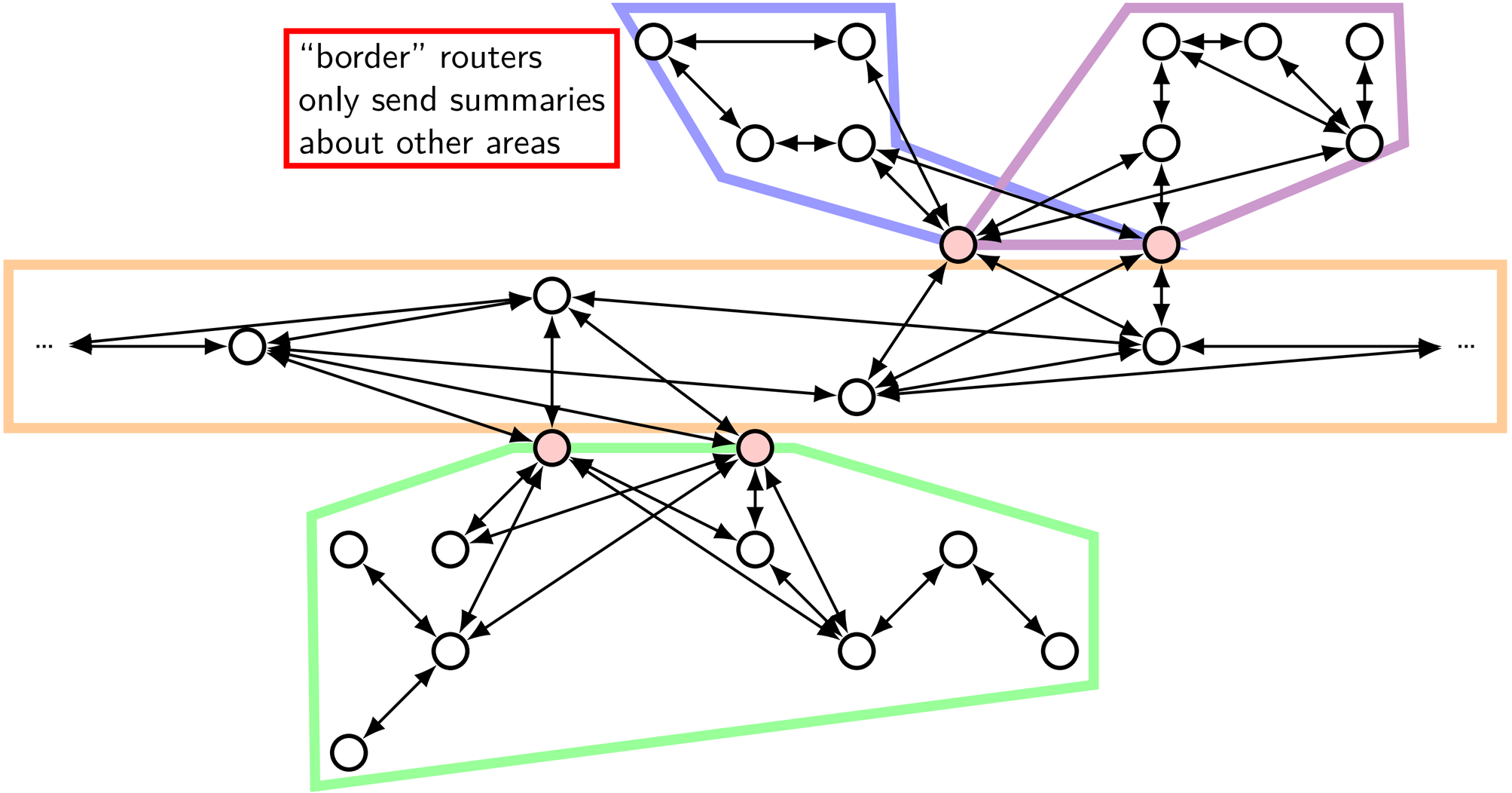
a big network



a big network

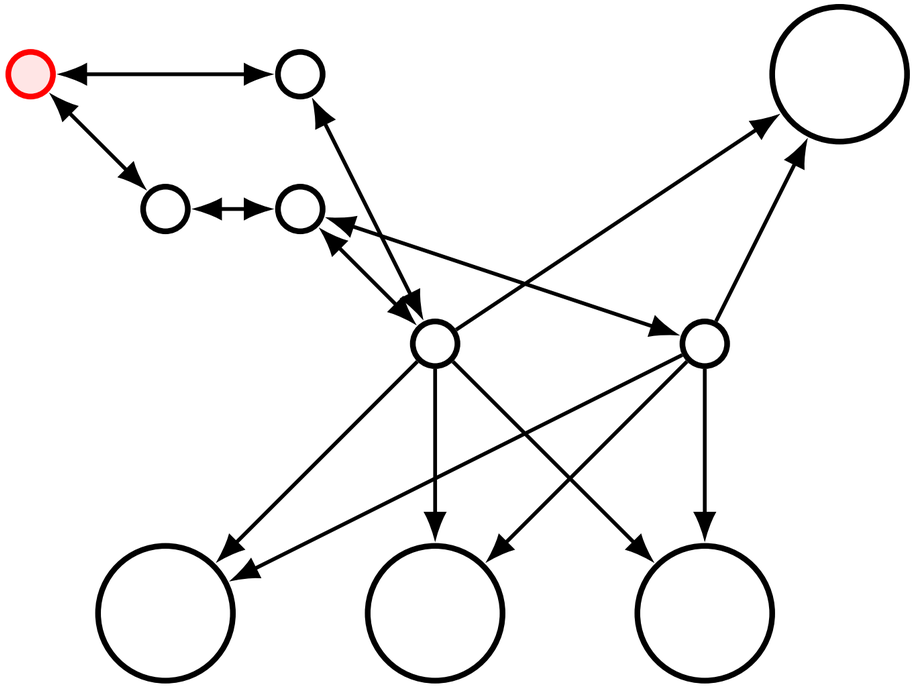


a big network



a big network

router sees simpler summary of network
→ hopefully faster routing



...

...

distance vector in link state?

summaries are distance vectors!

- area border routers just saying which networks + metric

idea: mix simpler distance vectors with more flexible link-state

but distance vector problems?

recall: count-to-infinity

let's say areas A, B, C, D all connected to each other…

… and area D goes offline:

could packet for D loop area A to B to C to A to B to C to …

OSPF solution: disallow this network configuration

backbone

OSPF area 0 is called “backbone”

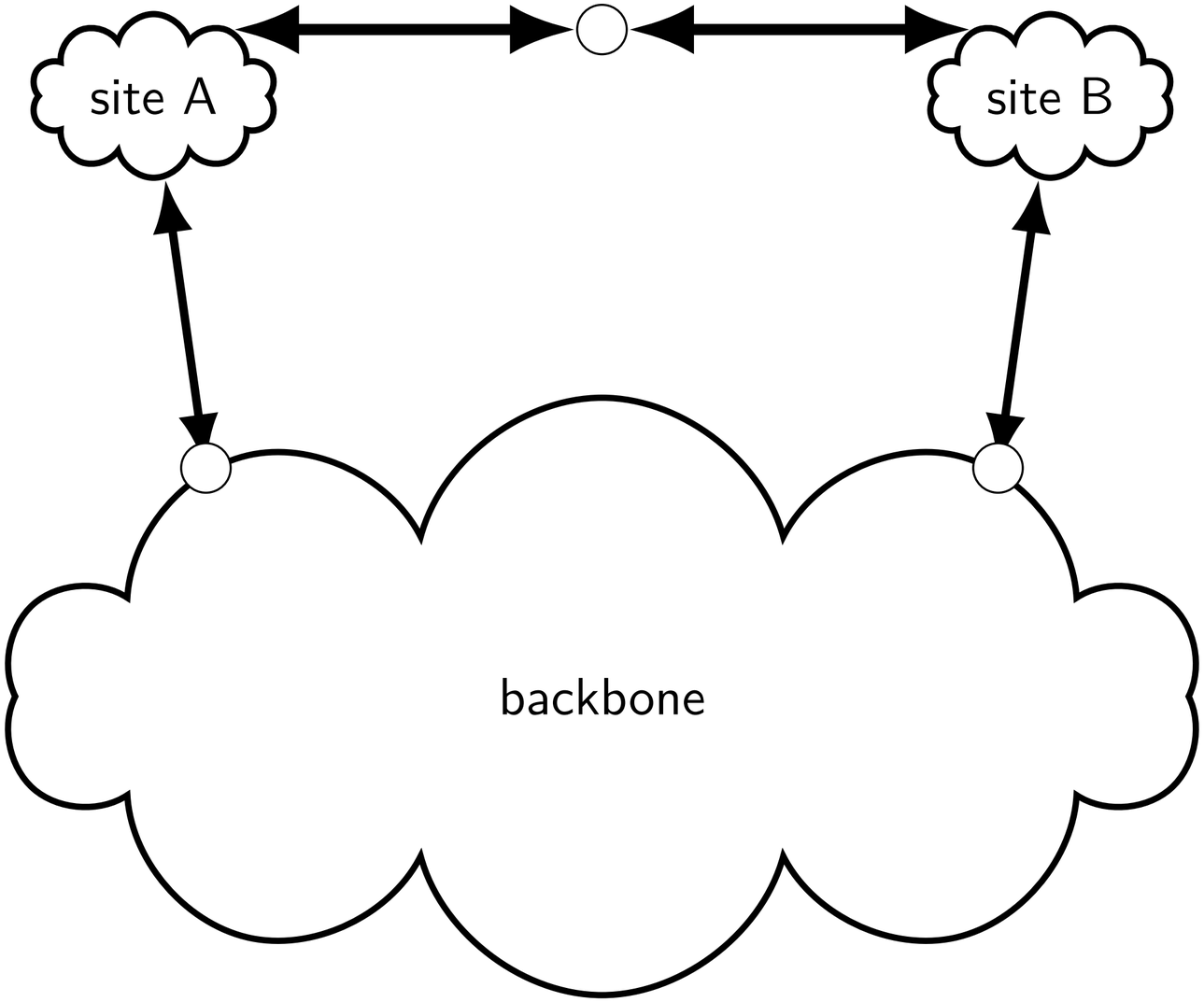
border routers only summarize routes sent to backbone *or* not obtained from other area border routers

means routing between areas must either:

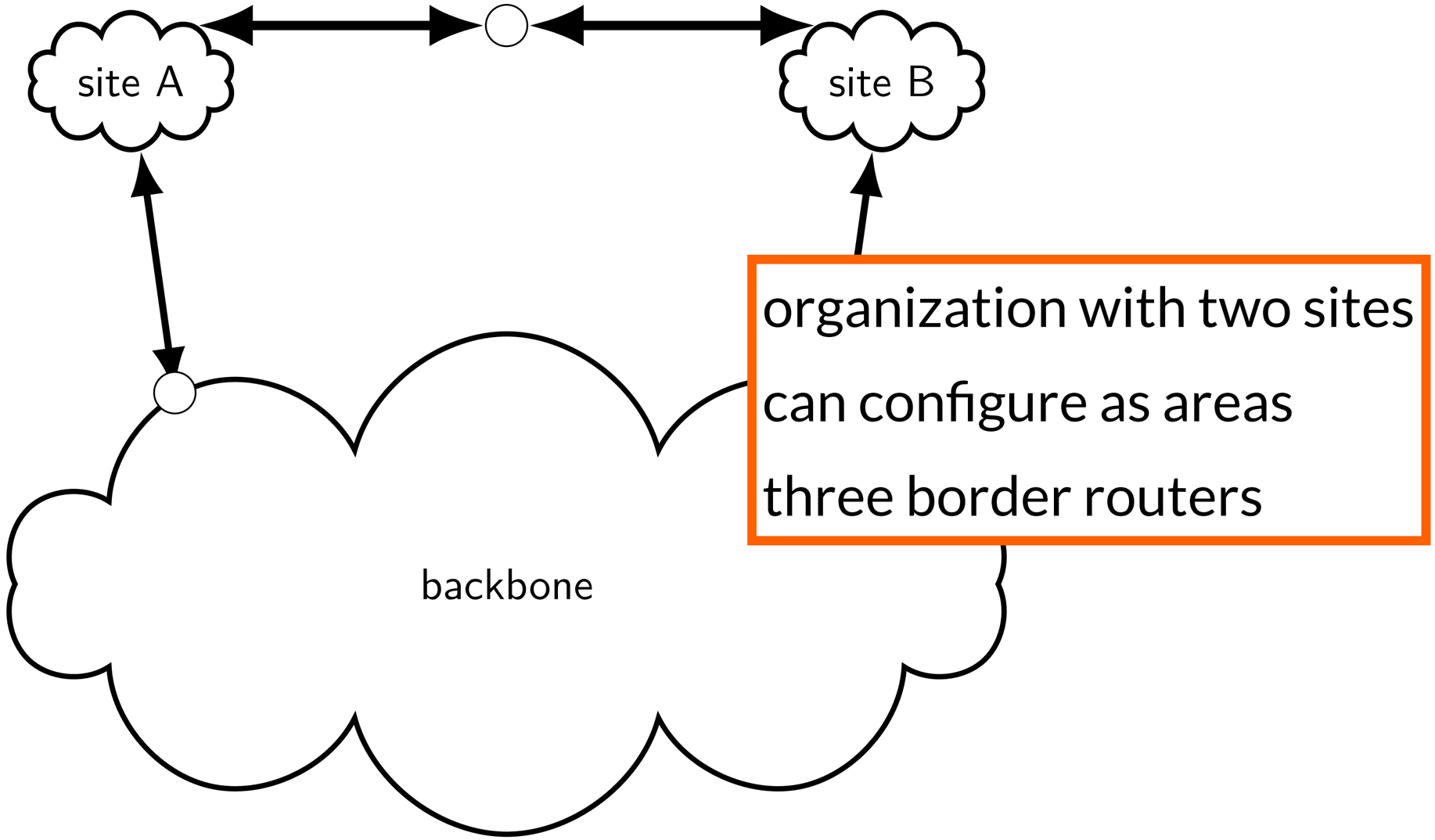
- go through the backbone, or
- only go through one border router

makes loops not possible

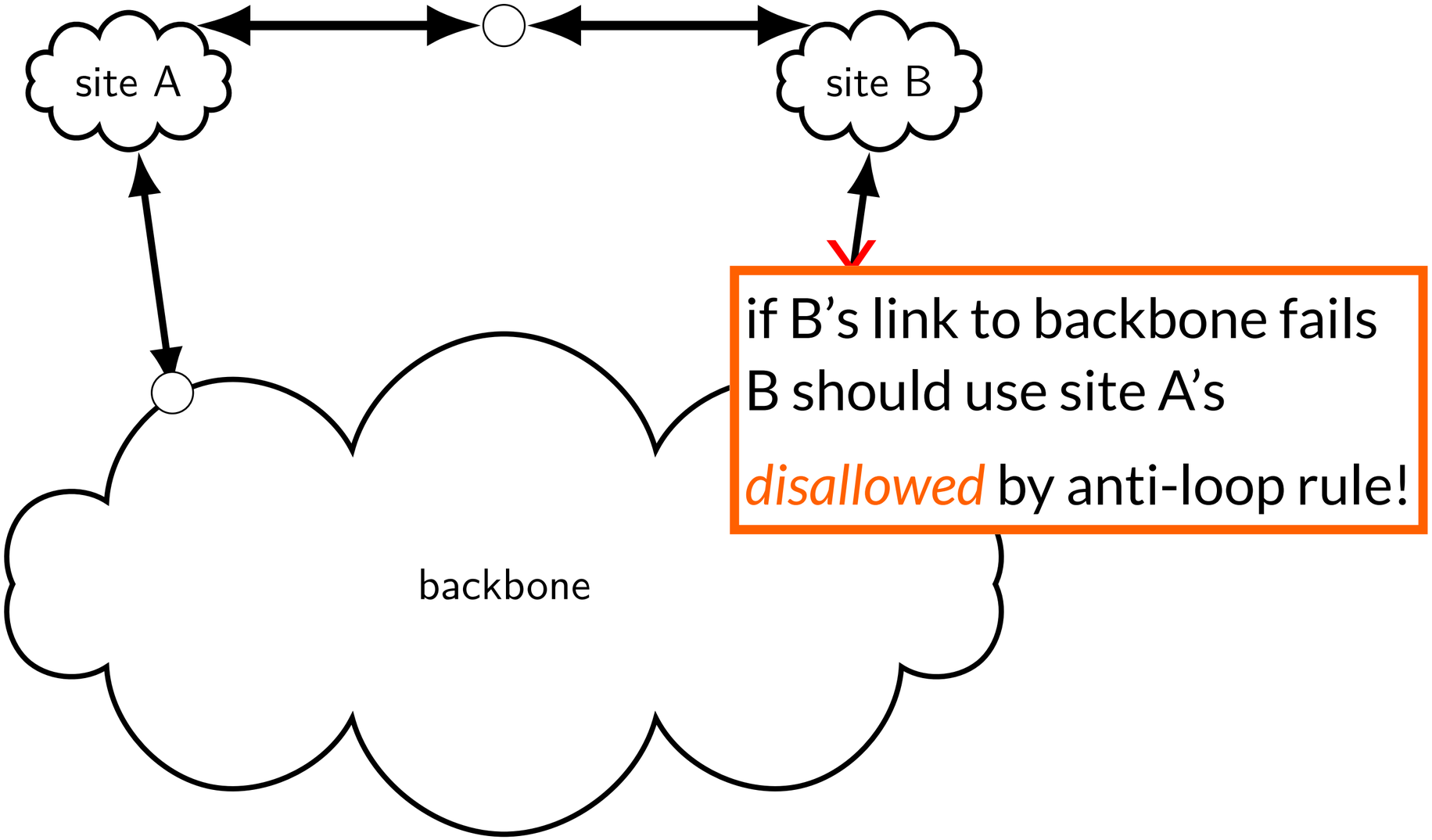
backbone limits



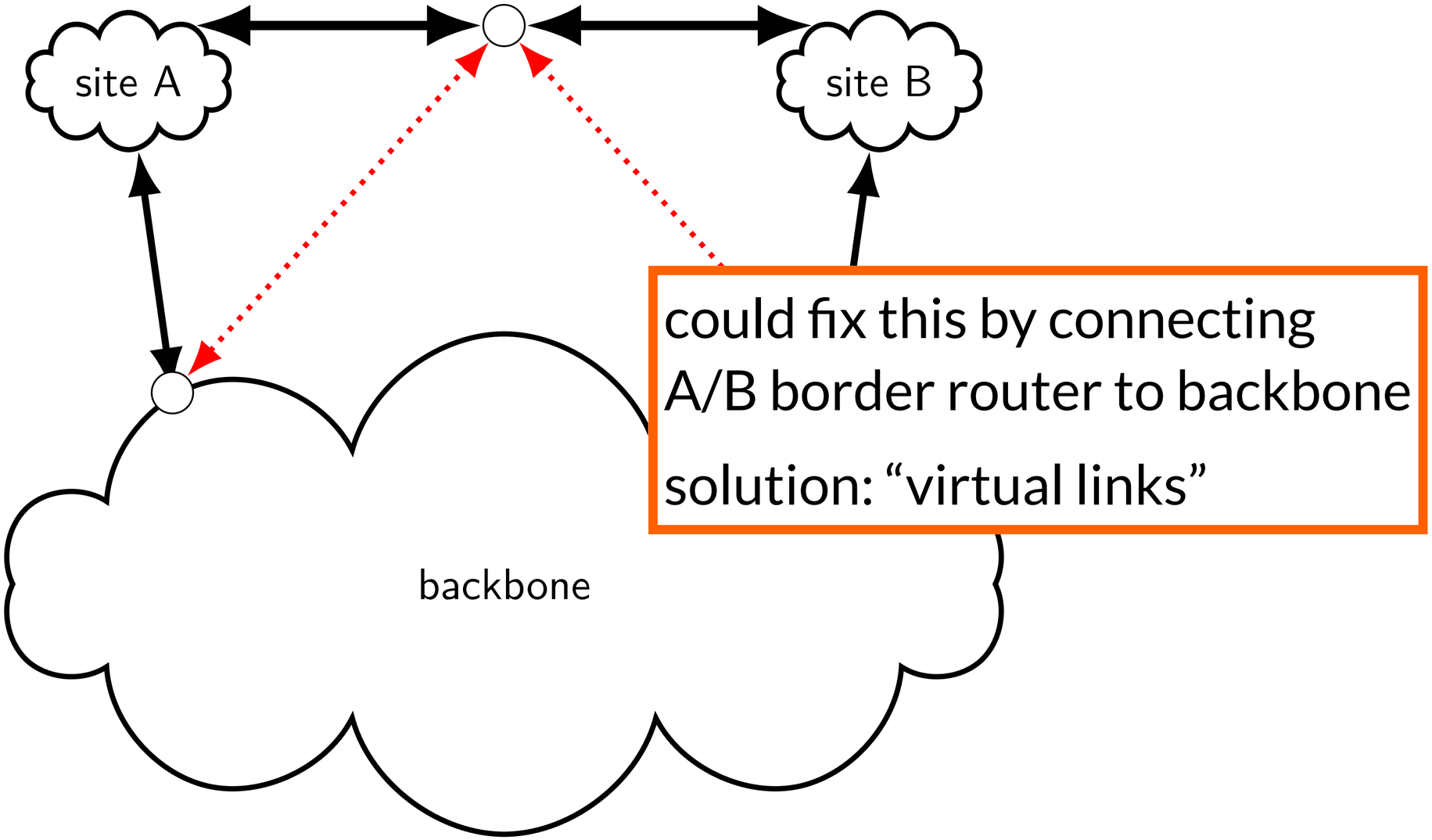
backbone limits



backbone limits



backbone limits



OSPF virtual links

“tunnel” backbone through another area

- route as if ‘direct’ connection between two border routers
- but connection implemented by going through area
- both ends considered part of backbone

metric for virtaul link = metric of route through area

configured explicitly by administrator

backup slides